Module – DevOps

Topic 1 - Introduction to DevOps

Level 5

What we will cover in this topic

- What is DevOps?
- DevOps Pipelines
- DORA
- Source Control

What is DevOps?

DevOps is a combination of:

Development (ie software engineering)

and

Operations (various activities typically undertaken by an IT department)

> “Devops isn't any single person's job. It's everyone's job”
>
> \- Robert Krohn
> HEAD OF ENGINEERING, DEVOPS AT ATLASSIAN

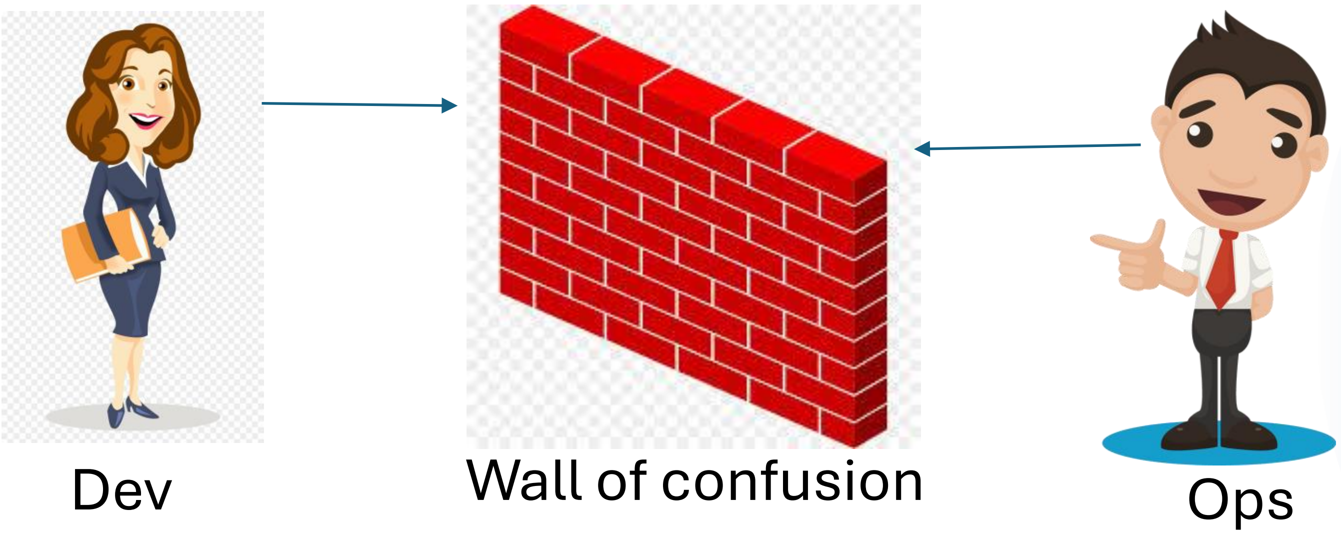
What is DevOps?

DevOps means merging the activities typically expected of Operations and merging it with Development.

This may mean having teams with both sets of people, or

It may mean having people with both sets of skills.

This reduces the silo effect.



What is DevOps?

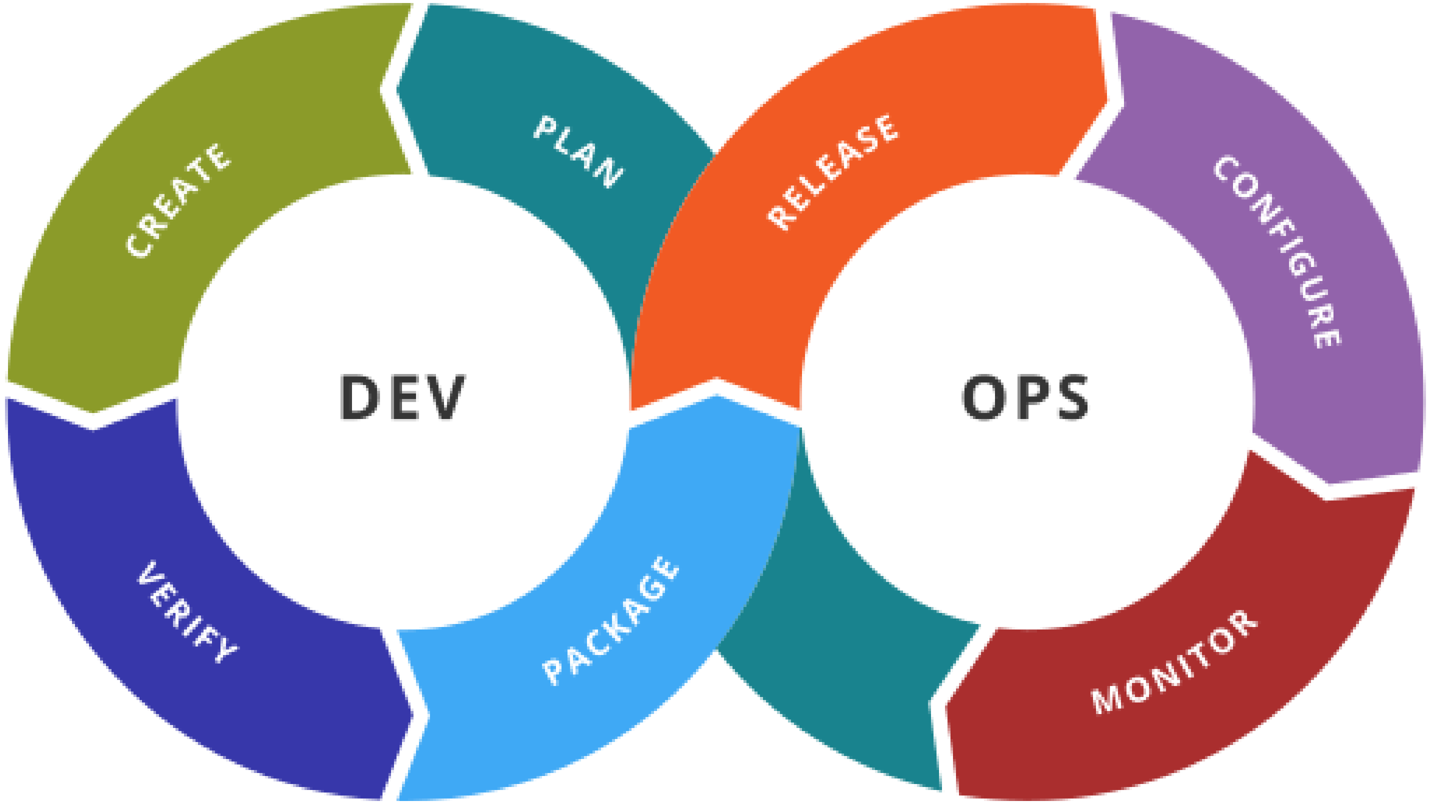
DevOps as a concept works on agile principles

- cross-functional teams
- continuous improvement
- shared ownership
- rapid feedback
- reduce waste
- increase value

There is an emphasis on automation and productivity

What is DevOps?

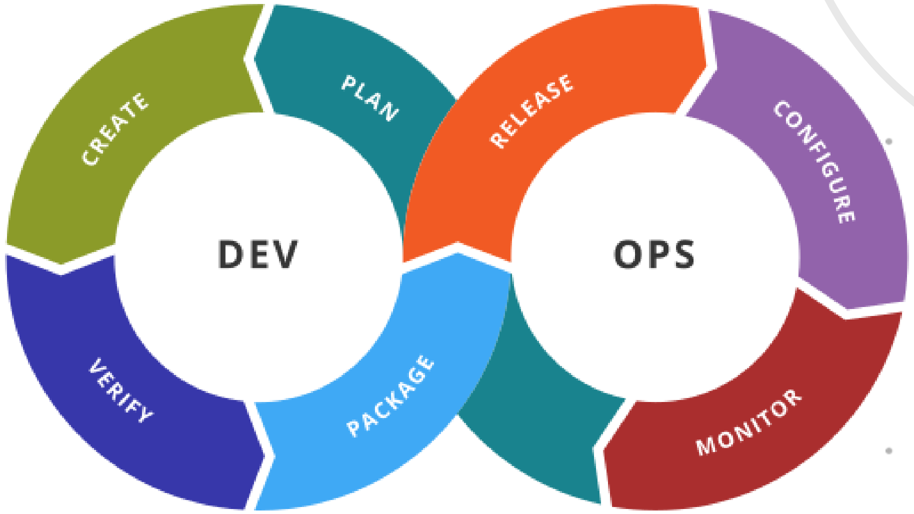
The DevOps Lifecycle



What is DevOps?

Notice the additional steps not normally associated with a development lifecycle

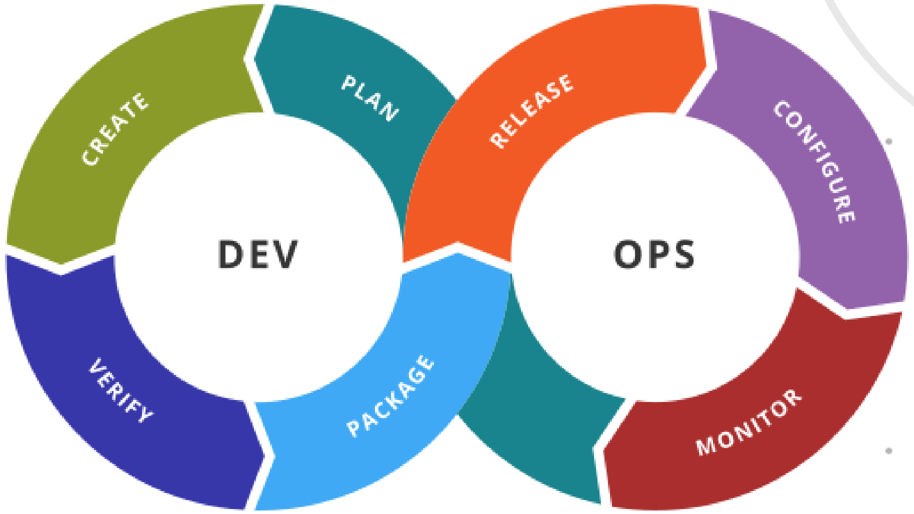
- (Package)
- Release
- Configure
- Monitor



What is DevOps?

Package

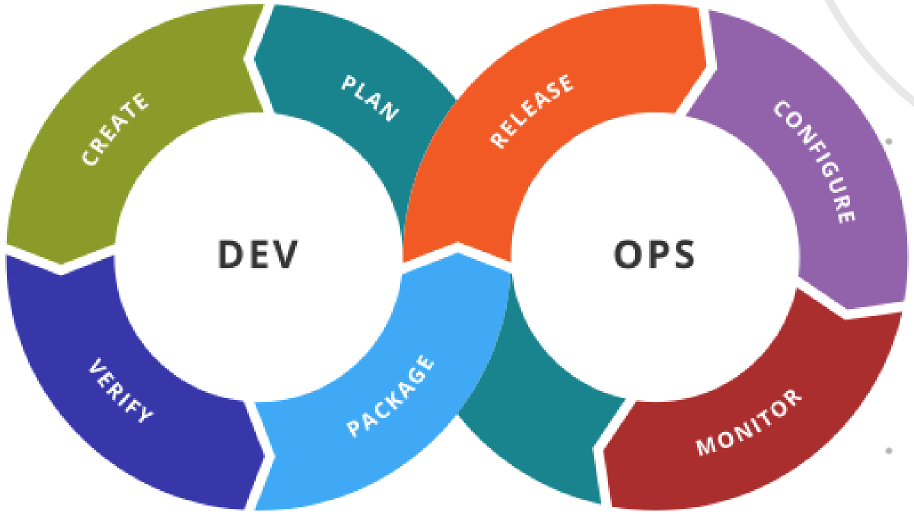
- Packaging is like zipping files
- It means packing all the needed software files into a 'package' so that it now the releasable software



What is DevOps?

Release

- Release is the process of placing the packaged software onto the machine(s) where it will be run

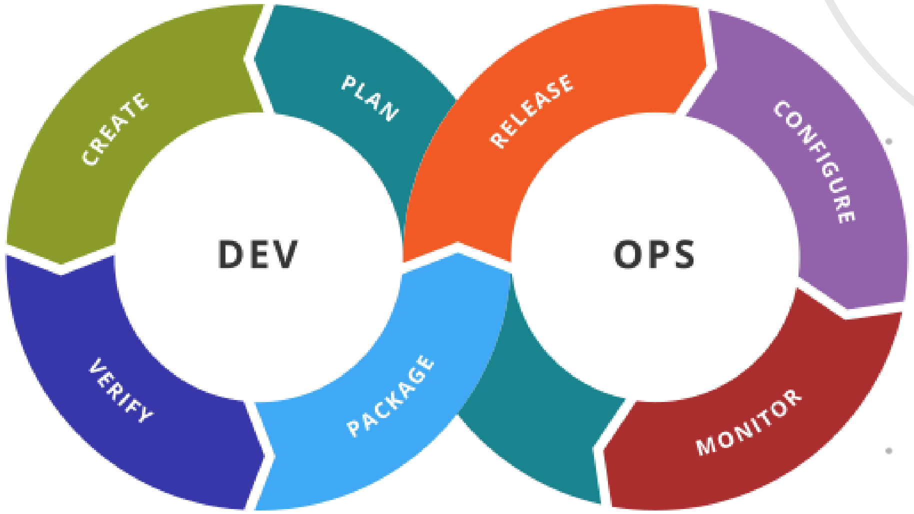


What is DevOps?

Configure

- Configuration is the process of knowing what software (and hardware) is being used for which purposes

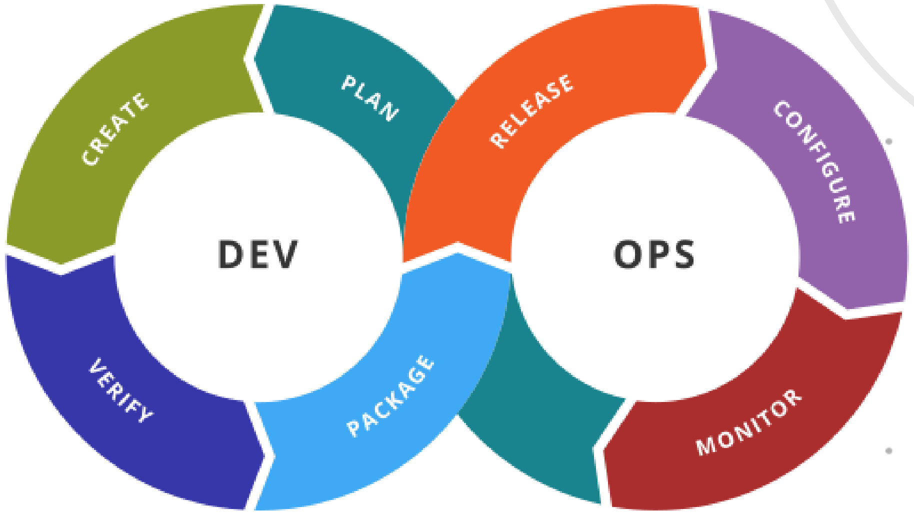
- It can also mean the properties of something (in this case a piece of software)



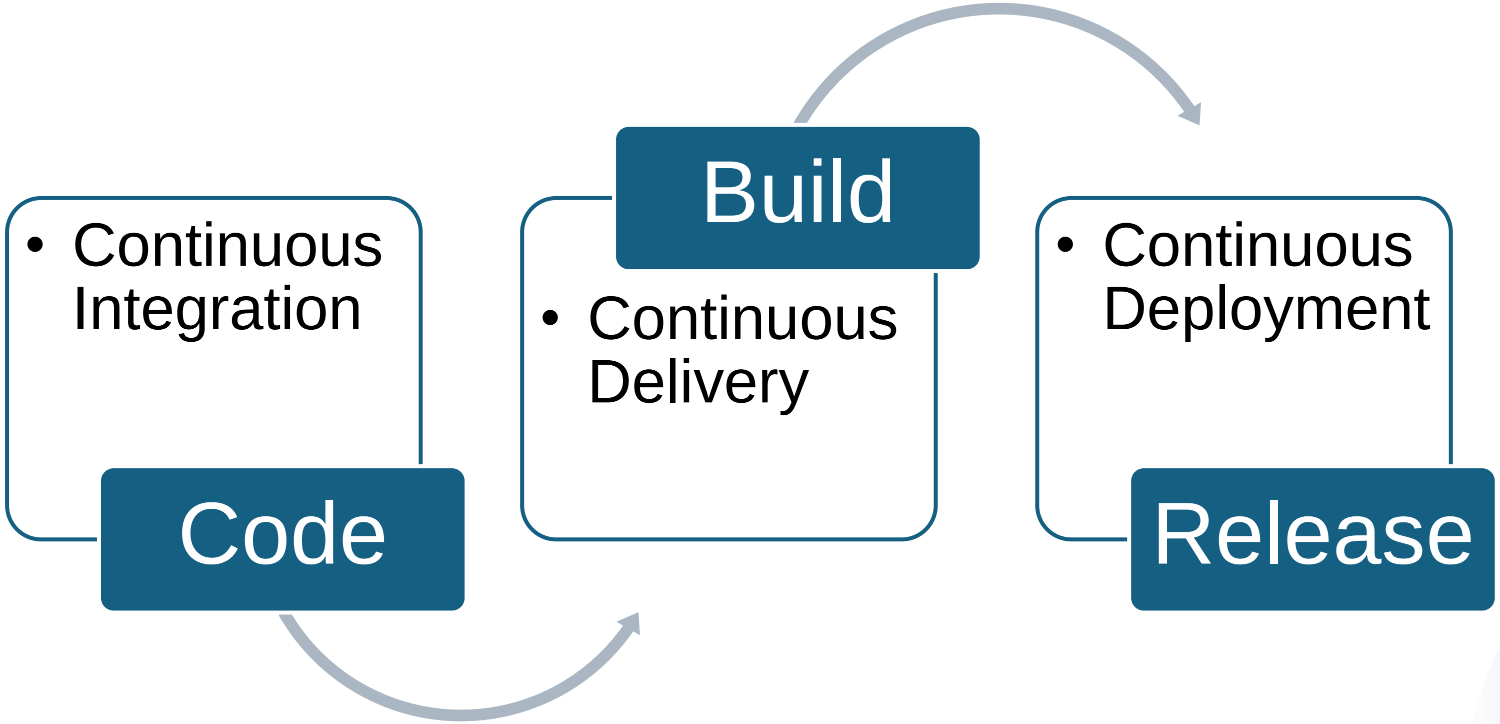
What is DevOps?

Monitor

- Monitoring is the process of checking that everything is working as expected in a proactive way (rather than waiting for it go wrong)
- For example regularly checking that the software is responding to requests



DevOps Pipelines



DevOps Pipelines

Benefits
• Reduced risk • Shorter review time • Better code quality • Faster bug fixes • Measurable progress • Faster feedback loops • Increased collaboration

BPP

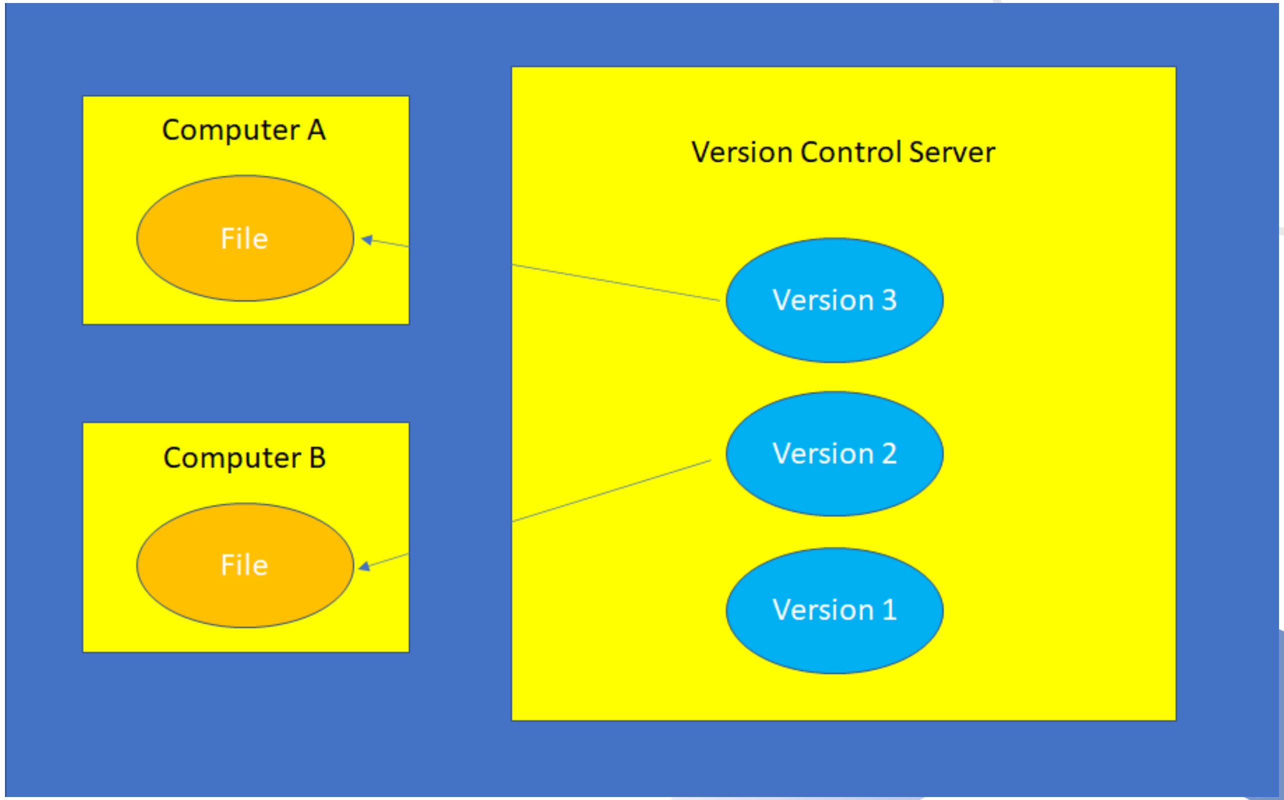
DORA

1. Deployment Frequency
2. Lead Time for Changes
3. Change Failure Rate
4. Mean Time to Restore (MTTR)

Source Control

Source control, aka version control, is a way of tracking changes to software, or really, any files.

It maintains a history over time that means you can see, and resurrect, previous versions of the file.



Source Control

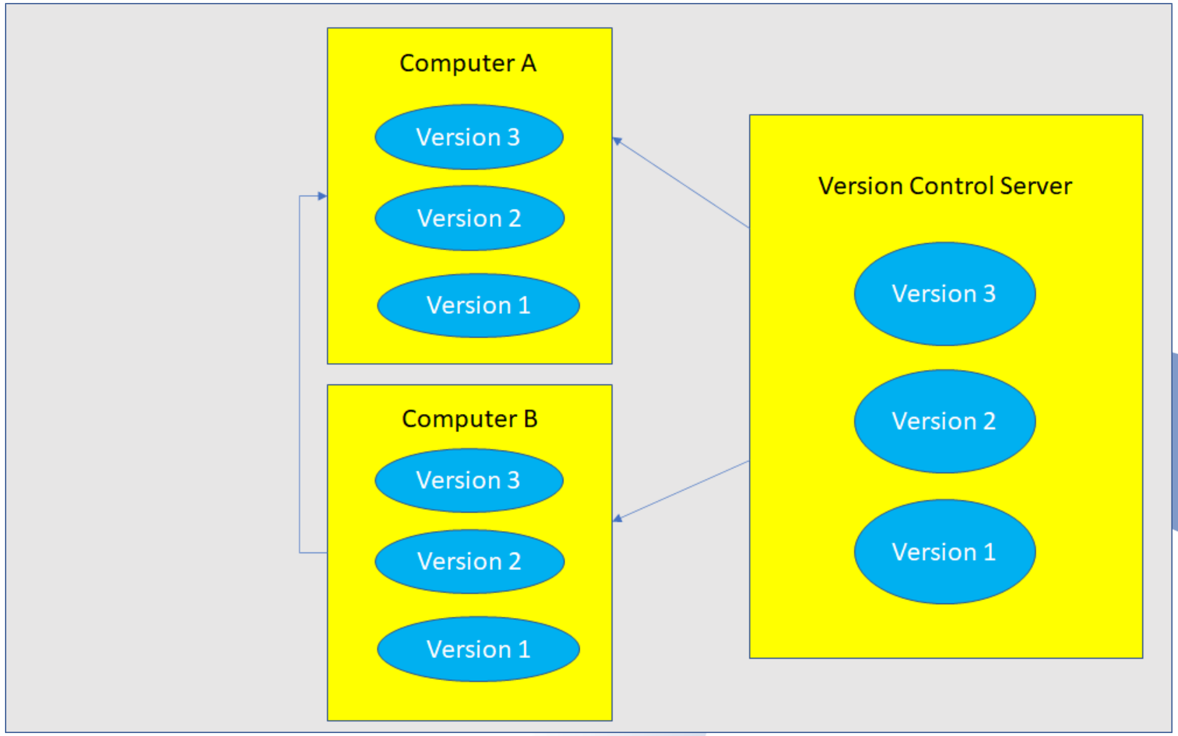
Source control also provides the ability for software engineers to work on the same piece of code at the same time without impacting the other engineer’s work.

Similarly, changes can be made to code that is being used in a released piece of software without impacting it, with the changes been added later when they are known to be working

Git

Git is a distributed source control system, invented by Linus Torvalds the inventor of the Linux operating system

A distributed source control system has the ability to have multiple ‘central’ versions of a file, and also allow anyone using that file to have multiple versions on their computer.



Source Control Terminology

Check Out

Check Out means to retrieve a file or set of files, of a certain version, to a local computer from the central source control server. Also called PULLing.

Check In

Check In is the opposite of Check Out. So sending a local copy, usually with changes, so a central source control server. Also called PUSHing.

Commit

Commit means to add your changed code to be ready for checking in. You would normally check in 1 or more pieces of code before pushing it.

Source Control Terminology

Branch

A branch is a copy of a set of files that means they can be worked on independently whilst maintaining the original.

Master

Master, or Master Branch, is the combined, central copy of the files. A branch would be taken from this (or another branch). Sometimes called the ‘trunk’.

Source Control Terminology

Working Directory

The Working Directory is your local copy of the files, pulled from a branch or master.

Staging Area

The Staging Area is a local copy of changes files, before they are committed.

Source Control Terminology

Repository

A repository is a central set of files. Usually related to each other, for example a piece of software's source code. Sometimes just called a repo.

Local Repository

A local repository is a copy of the repository on a local computer. Usually a copy of the master/branch from the server.

Remote Repository

A remote repository is the repository on a computer other than your own.

BPP

Source Control Terminology

Merge

Merging is combining code between a local repository and a remote one, or combining code in 2 branches.

Clone

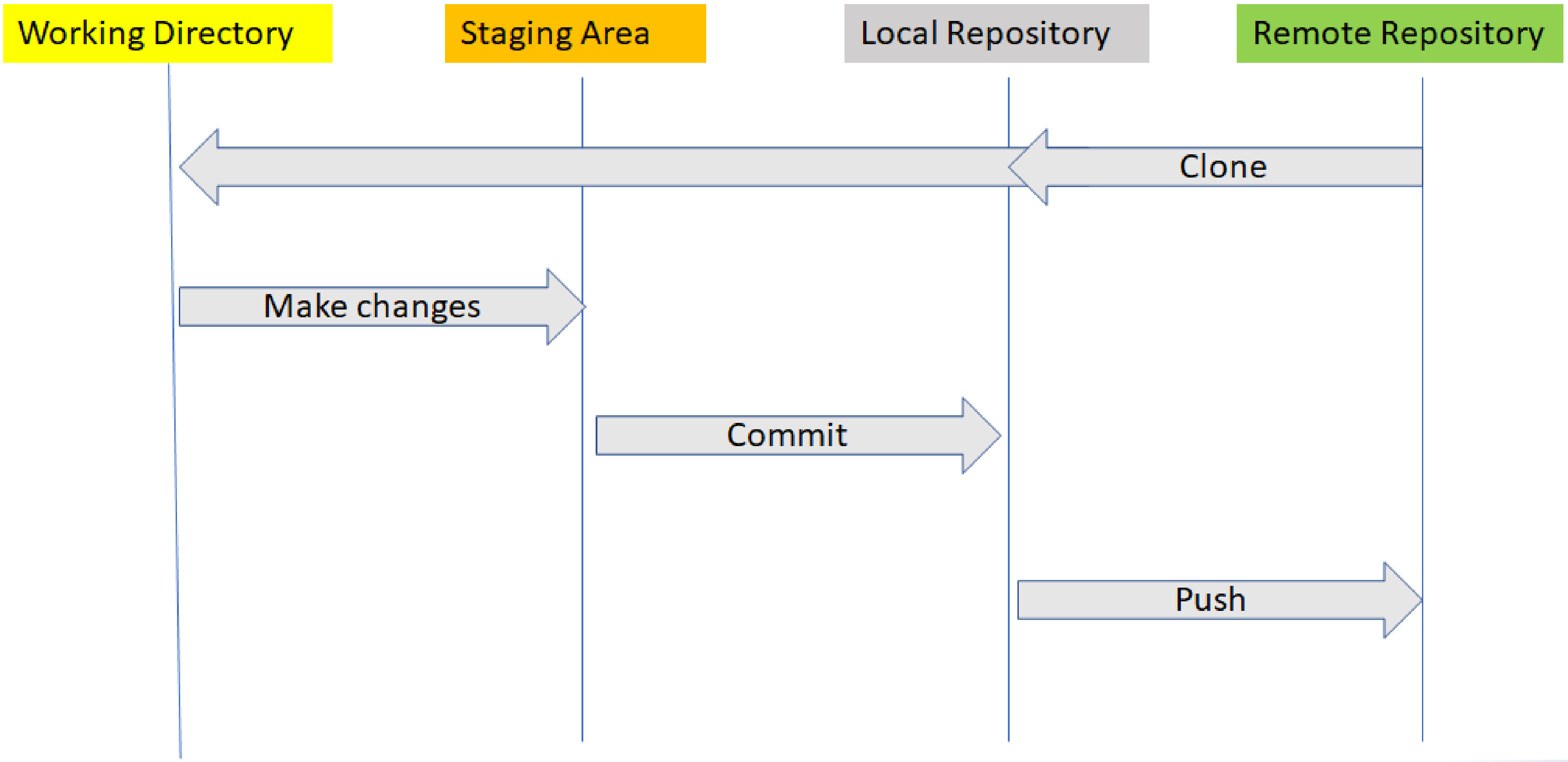
Cloning is a complete copy of a remote repository. You create a local repository by cloning a remote one.

Tag

Tagging is basically applying a name to a version of the code so that it can be resurrected or located when needed. Especially useful for tagging the state of software when it is released or for versioning entire projects.

BPP

Source Control Workflow



Summary

- What is DevOps?
- DevOps Pipelines
- DORA
- Source Control

DevOps
Topic 1

DevOps Concepts